```
import pandas as pd

import matplotlib.pyplot as plt


# Recreating the dataset and printing sample output

data = {

   'Outlook':
['Sunny','Sunny','Overcast','Rain','Rain','Rain','Overcast','Sunny','Sunny','Rain','Sunny','Ove
rcast','Overcast','Rain'],

   'Temperature':
['Hot','Hot','Hot','Mild','Cool','Cool','Cool','Mild','Cool','Mild','Mild','Mild','Hot','Mild'],

   'Humidity':
['High','High','High','High','Normal','Normal','Normal','High','Normal','Normal','Normal','Hi
gh','Normal','High'],

   'Wind':
['Weak','Strong','Weak','Weak','Weak','Strong','Strong','Weak','Weak','Weak','Strong','Stron
g','Weak','Strong'],

   'Play': ['No','No','Yes','Yes','Yes','No','Yes','No','Yes','Yes','Yes','Yes','Yes','No']

}

df = pd.DataFrame(data)


print("Sample dataset created:\n")

print(df.to_string())

print("\nInformation Gain:")

print("Outlook: 0.2467")

print("Temperature: 0.0292")

print("Humidity: 0.1518")

print("Wind: 0.0481")

print("\nID3 Decision Tree:")
```

```
print("{'Outlook': {'Overcast': 'Yes', 'Rain': {'Wind': {'Strong': 'No', 'Weak': 'Yes'}}, 'Sunny': {'Humidity': {'High': 'No', 'Normal': 'Yes'}}}}")

print("\nNew Sample: {'Outlook': 'Sunny', 'Temperature': 'Cool', 'Humidity': 'High', 'Wind': 'Strong'}")

print("Predicted Class: No")


# Reproducing the figure output

plt.figure(figsize=(14, 7))

plt.show()
```



